

ARLINGTON, WA, USA

WMO: 727945

Lat: 48.161N

Lon: 122.159W

Elev: 137

StdP: 14.62

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 04205

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.4	23.6	13.6	11.0	23.7	18.4	13.9	26.5	26.5	49.8	24.0	49.3	1.2	0	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.7	83.4	66.7	80.6	64.9	76.6	63.0	67.4	81.2	65.6	78.4	64.0	75.5	9.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	82.0	71.9	60.7	79.7	71.5	58.5	73.7	69.8	31.7	81.1	30.4	78.8	29.1	75.2	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.1	15.6	DB	14.8	89.1	5.6	4.0	10.8	92.0	7.5	94.3	4.4	96.6	0.3	99.5	(4)
(5)			WB	13.7	70.7	5.7	2.2	9.6	72.3	6.3	73.5	3.1	74.7	-1.1	76.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	40.5	40.6	44.6	48.7	53.9	58.5	63.1	63.5	58.8	50.3	43.4	38.5	(6)	
	DBStd	10.18	6.82	5.94	5.27	4.65	5.07	4.50	4.31	3.93	4.53	5.19	6.39	6.48	(7)	
	HDD50	1466	298	264	180	76	18	1	0	0	1	60	210	358	(8)	
	HDD65	5388	758	682	631	489	345	202	89	76	192	456	647	821	(9)	
	CDD50	1627	5	2	14	37	139	256	406	419	264	70	13	2	(10)	
	CDD65	73	0	0	0	0	1	7	30	30	5	0	0	0	(11)	
	CDH74	1314	0	0	2	10	41	145	481	484	150	1	0	0	(12)	
	CDH80	301	0	0	0	1	5	30	134	108	23	0	0	0	(13)	
(14) Wind	WSAvg	5.1	5.4	5.1	6.4	5.8	4.9	4.9	4.9	4.5	4.2	4.5	5.4	5.0	(14)	
(15) Precipitation	PrecAvg	46.7	5.4	3.7	4.5	3.9	3.5	2.7	1.2	1.2	2.4	4.7	7.0	6.3	(15)	
	PrecMax	57.4	9.7	7.1	7.3	6.0	6.1	5.2	3.5	4.1	5.0	8.4	12.0	9.6	(16)	
	PrecMin	37.0	2.8	0.6	1.8	1.3	1.2	0.7	0.0	0.0	0.7	1.4	2.7	3.5	(17)	
	PrecStd	6.2	1.6	1.6	1.3	1.0	1.2	1.3	0.8	1.1	1.3	2.1	2.4	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	58.9	67.4	73.4	79.2	83.9	89.9	87.6	82.5	70.5	61.5	56.8	(19)	
		MCWB	52.1	49.4	53.9	59.4	63.1	67.4	69.0	67.3	65.0	58.2	55.7	52.3	(20)	
	2%	DB	54.7	54.9	61.2	66.2	72.9	79.0	83.6	82.3	78.9	65.8	57.2	52.5	(21)	
		MCWB	50.2	47.8	51.3	55.5	60.5	64.4	67.0	66.1	63.3	56.9	52.7	49.0	(22)	
	5%	DB	52.7	52.3	57.0	62.8	69.7	73.2	79.3	79.5	73.4	63.0	54.9	50.1	(23)	
		MCWB	48.8	46.7	49.2	52.9	59.1	61.1	64.4	64.5	61.2	55.5	50.9	47.0	(24)	
	10%	DB	50.2	49.7	54.3	58.0	64.3	70.1	75.1	75.3	70.3	59.4	53.6	47.7	(25)	
		MCWB	46.8	45.7	48.1	50.0	55.9	59.8	62.9	63.0	60.0	53.3	50.1	45.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	51.8	55.4	60.6	64.3	68.5	70.4	69.1	67.1	61.5	57.3	53.1	(27)	
		MCDB	56.3	56.0	65.0	71.6	75.7	82.2	85.6	83.0	79.6	66.9	61.1	55.8	(28)	
	2%	WB	50.9	49.2	52.6	56.1	61.8	64.9	67.4	67.0	64.5	58.1	53.7	49.4	(29)	
		MCDB	53.8	53.2	59.3	64.0	72.8	77.7	81.5	80.8	76.2	63.6	56.6	52.0	(30)	
	5%	WB	48.7	47.4	50.6	53.4	59.4	62.1	65.4	65.5	62.5	56.0	51.6	46.8	(31)	
		MCDB	51.7	51.1	56.1	60.6	68.0	72.5	78.1	78.4	73.1	61.5	54.8	48.9	(32)	
	10%	WB	46.5	45.8	48.5	51.1	56.9	59.9	63.5	63.7	60.6	54.1	49.8	45.4	(33)	
		MCDB	49.0	48.8	53.6	57.4	64.3	68.2	74.5	75.0	69.0	58.6	52.8	47.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	14.7	16.0	17.2	18.1	18.5	21.9	22.7	21.1	16.4	13.5	11.6	(35)	
		MCDBR	13.6	19.4	22.6	26.1	27.3	27.7	30.1	30.2	29.1	21.9	14.7	11.9	(36)	
		MCWBR	10.1	12.5	14.5	16.3	16.5	16.0	15.8	15.7	16.7	14.7	10.8	9.2	(37)	
	5% WB	MCDBR	12.0	14.7	19.6	22.2	24.7	25.8	27.8	27.8	26.9	18.7	12.2	10.5	(38)	
		MCWBR	9.4	10.7	13.1	14.6	15.5	15.3	15.5	15.0	15.8	13.3	9.6	8.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.311	0.307	0.320	0.347	0.368	0.355	0.346	0.350	0.336	0.325	0.311	0.313	(40)	
	taud		2.525	2.546	2.538	2.451	2.414	2.468	2.514	2.513	2.546	2.572	2.559	2.466	(41)	
	Ebn at Noon		245	274	287	287	284	288	289	283	278	263	243	226	(42)	
	Edh at Noon		19	23	27	33	36	34	32	30	27	23	19	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		312	590	902	1351	1704	1872	2092	1773	1270	712	364	250	(44)	
	RadStd		44	75	104	126	138	178	153	117	106	82	52	35	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page